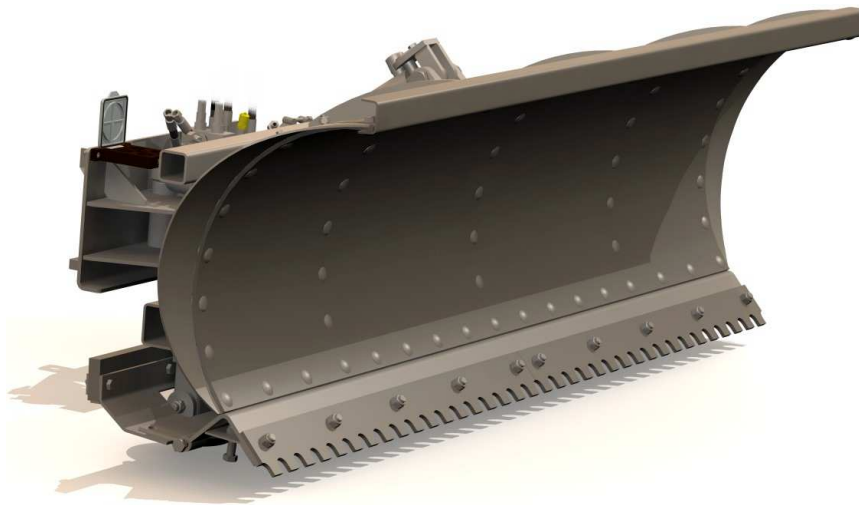




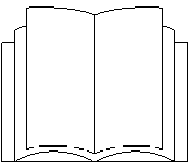
Operator's Manual for Attachment



Snow Blade 2200

Product number:

300-700-series loaders:
A21493



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

AVANT
AVANT TECNO OY
e-mail: sales@avanttecno.com

Ylötie 1
33470 Ylöjärvi
FINLAND
Tel. +358 3 347 8800
Fax +358 3 348 5511

www.avanttecno.com

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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or property damage:

	DANGER:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.



Note:

This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment, particularly the use of the equipment on road traffic. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader and attachment and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost or damaged, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The AVANT Snow blade is an attachment suitable for use with AVANT 600 and 700 series multi purpose loaders. Any special options for the loader are not necessary, but the telescopic boom and the electric joystick are recommended. The blade is equipped with hydraulic side slewing as standard, and thanks to the symmetric design, snow can be moved efficiently to either side of the blade. The edge of the blade is made from a wear resistant cutting profile, which removes compacted snow effectively. A rubber blade is also available for more sensitive surfaces. The blade

The shape of the blade is designed to throw snow further away than a straight plow, which makes clearing large areas and roads quicker. Also the sideways force created by the snow blade is less than what is caused by a straight snow plow, and this allows to move larger amounts of snow at once. In addition to the floating attachment bracket, the blade is also equipped with a hydraulic blade release system. The release system allows the blade to tilt backwards if an obstacle is hit, preventing sudden stopping of the loader and protecting both the driver and the blade.

The attachment has been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. Because of the pressure accumulator installed to the separate hydraulic system of the blade tilting, only a person experienced in repairments of hydraulic system is allowed to do any maintenance or service to the hydraulic components of the blade. Appropriate protective equipment and original spare parts must be used for the maintenance. Familiarise yourself with the manual's instructions regarding service and maintenance. Please contact your AVANT retailer or nearest service point if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for using the Snow Plow

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards. The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- When attaching the dozer blade to the loader, ensure that the locking pins of the loader's quick attach plate are **in the lower position** and that they have **locked the attachment to the loader**.



- Do not drive at high speed unless you can be sure that there are no obstacles at the working area. The loader may stop suddenly as it hits an obstacle regardless of the blade release mechanism. Remember that there may be hidden obstructions or fragile surfaces under the snow.
- Pay attention to the surroundings and any other persons and machines moving in the vicinity. Pay attention to the contours of the terrain as well as other hazards, such as branches and trees reaching the driver's area.
- On slippery areas drive calmly and deliberately.
- Never leave the driver's seat if the blade is lifted from the ground. Lower the boom and shut down the loader engine before any adjustments or maintenance work.
- Do not modify the structure of the attachment in a way that would affect safety.
- Also read the safety instructions for the loader in the loader manual.



Note that the loader boom moves down and that the attachment may be tilted even if the engine has been stopped. Never go under a raised boom or an attachment.



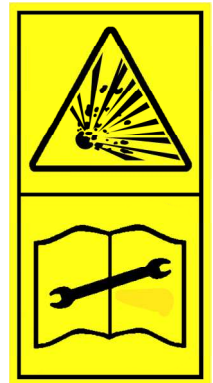
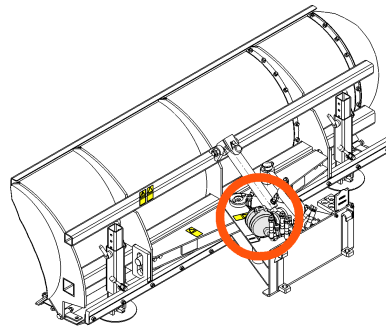
Familiarise yourself with the loader controls in a safe area. Pay particular attention to safe stopping of the equipment. Practice driving on slippery areas. Snow chains may be necessary.



Note that driving characteristics of the loader may be altered on slippery surfaces and cold conditions. Allow the loader to warm up before use.



The hydraulic circuit of the blade yielding system includes a pressure accumulator. Due to this, only a professional experienced in hydraulic systems is allowed to service or repair the hydraulic system of the snow blade. Mishandling or careless service of the hydraulic system and the pressure accumulator may lead to sudden burst of hydraulic oil or compressed gas and serious injuries. Read the instructions regarding service from chapter 7.



Before handling hydraulic components, make sure that the hydraulic system of the attachment and the loader are completely depressurized. Never handle pressurized fittings. High pressure fluid ejection may penetrate skin and cause serious injuries. Seek medical attention immediately in case hydraulic fluid is injected into the skin.



Remember to wear proper protective clothing:



- The noise level at the driver's seat may exceed 85 dB(A) depending on loader model and working conditions. Wear hearing protection while working with the loader.
- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic components.

4. Snow Plow specifications

Listed below are the labels and markings, which must be visible on the equipment. Replace any warning label, that has become unclear or has detached completely. New labels are available via your retailer or contact information provided on the cover.



The warning labels contain important safety information. Replace damaged or missing warning labels with new ones.



A46771



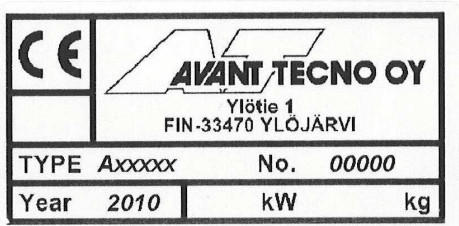
A46772



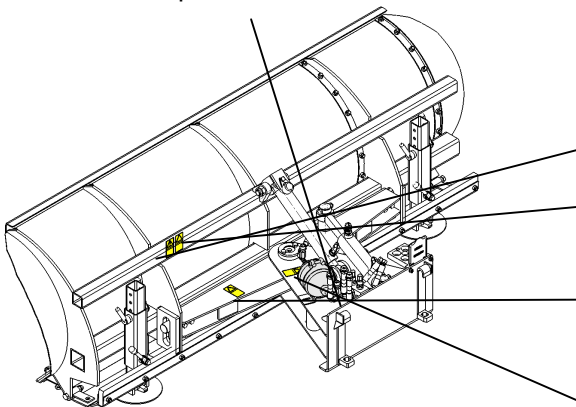
A46797



A414631



Identification plate of the attachment



Warning decal locations

Table 1 – Decal warning messages

Decal	Explanation
A46771	Read instructions before use.
A46772	Do not go under a raised attachment or the loader boom, keep a safe distance to lifted equipment.
A46803	Crushing hazard, keep clear of moving parts, floating bracket and blade tilting.
A414631	Pressurized vessel in the hydraulic circuit, hazard of eruption or discharge of energy stored in pressure accumulator. Follow maintenance instructions.

Table 2 – Snow blade specifications

Product number:	A21493
Working width:	2 200 mm
Dimensions in mm, total (width / height / length):	2 500 / 750 / 1 150
Weight:	375 kg
Slewing angle:	±30°
Maximum allowed input of hydraulic energy:	22,5 MPa (225 bar) 60 l/min
Optional rubber blade assembly:	A21497
Suitable AVANT loaders:	600-700 series

Table 3 - Pressure accumulator specifications

Part number:	65606
Accumulator type	Diaphragm accumulator
Manufacturer and product number:	Olaer ELM 0,75-350/00/CG01125
Nominal volume	0,75 l
Gas precharge:	80 kPa (80 Bar), nitrogen, 20°C

5. Assembly of the Snow Plow

The snow blade is ready for use as delivered. Attach the blade and its hydraulic hoses to the loader as instructed. Read the instructions related to using the equipment and begin the practise its use at a safe area.

5.1. Attaching the snow plow to the loader

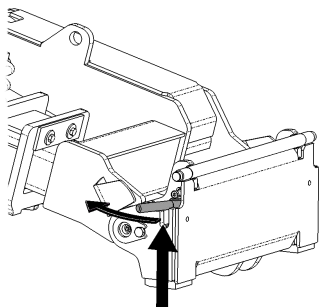
Mounting the attachment to the loader is quick and easy, but it must be done carefully. An attachment that has not been locked may come off the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised to a height of over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3 and the instructions provided in the loader manual.

The attachment is mounted to the loader boom by using the quick attach plate and the counterpart on the attachment. The locking on the loaders is semi-automatic, but it is recommended to lock the locking pins manually.



Never move or raise an attachment that has not been locked.

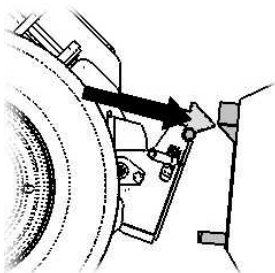
The snow blade is attached to the loader as follows:



Step 1:

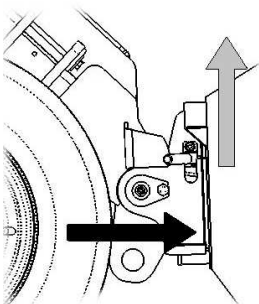
- Lift the quick attach plate locking pins up and turn them backwards into the slots so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.

Step 2:



- Turn the quick attach plate hydraulically to an obliquely forward position. The angle between the loader attachment plate and the attachment coupling should be about as shown in the adjacent picture.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.

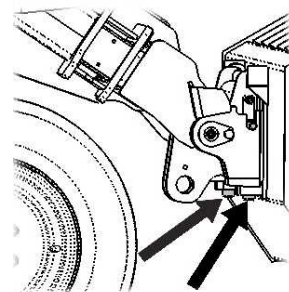
Step 3:



- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Lock the locking pins manually as you connect the hydraulic fittings.



Before moving or raising the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides. Unlocked attachment may fall on the boom or operator and cause injuries.



Never use the automatic locking of the locking pins **at a height of over half a metre**. The floating mechanism of the attachment may allow the attachment to move unexpectedly. We recommend that you lower the attachment on the ground and secure the locking manually when connecting the hydraulic fittings.

5.2. Connecting and disconnecting the hydraulic couplings

The attachment is equipped with either standard quick couplings or the multiconnector system. If you want to change the type of the hydraulic couplings, contact your Avant dealer or service point for instructions or installation services.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting standard quick couplings:

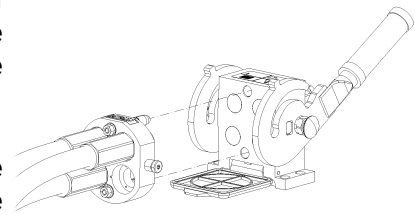
Before connecting or disconnecting the standard quick couplings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times. This way, there will be no residual pressure in the system and it is possible to connect the fittings.

To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the coupling equipped with a coloured cap is connected to the corresponding coupling of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.



The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Shut down the loader and release the residual hydraulic pressure.

Disconnecting the hydraulics:

Before disconnecting the fittings, lower the attachment on solid and level surface. Turn the control levers of the auxiliary hydraulics to its neutral position.



When removing the attachment, always disconnect the hydraulic couplings before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Releasing residual hydraulic pressure:

In case residual pressure is left in the hydraulic system of the attachment, it is often possible to disconnect the hydraulic couplings, but it may be difficult to connect them the next time. If the fittings will not connect, the residual pressure must be released by turning the auxiliary hydraulics control lever of the loader, when the engine is turned off. To make sure that there will not be residual pressure in the hydraulic system of the side turning of the snow blade, shut down the loader engine and move the auxiliary hydraulics control lever of the loader before disconnecting the couplings.

6. Instructions for use

Check the attachment and the operating environment once more before starting to work. Quick inspection of the equipment and the operating area before use are parts of ensuring safety and the best performance of the equipment.

- Check that the attachment is intact and that all obstacles, including any possibly hidden ones, have been removed from the working area before operation. Remember correct working methods and avoid leaving the driver's seat unnecessarily.
- Check that the blade yielding mechanism works as intended. Ensure that there are no obstructions near the slewing hinge; check that the hydraulic hoses or the electric harness can not get squeezed between any components or that they are not exposed to stretch at any position.
- Operate the attachment only from the driver's seat.
- Fasten seat belt when working with snow blade. The loader may stop suddenly and unexpectedly when an obstacle is hit regardless of the yielding mechanism.

- Check regularly to ensure there are no hydraulic leaks. See chapter 7 for maintenance information.
- Ensure that the loader and the attachment are used in a safe and intended manner. Ensure that the operating area is illuminated well enough. Do not allow children to use the equipment.
- See the operator's manual of the loader for information of safe operation.

Check the operating area before using the equipment:



- Make sure, that there are no hidden obstacles, for example, under the snow. Running into an obstacle the machine may stop suddenly and unexpectedly regardless of the blade yielding system or the boom floating of the loader.
- Ensure that there is nobody else in the working area. Do not allow assisting persons to go directly in front of the snow plow or to reach the boom danger zone. Also ensure that it is safe to reverse with the loader.



Tilting the attachment too much up may cause the blade to tip over on the loader boom. To prevent this, always keep the snow plow near the ground.

Turning the blade:

The blade slewing is controlled by either auxiliary hydraulics control lever of the loader or the electric switches of the joystick. One-pump setting is normally adequate when operating with the blade.

Blade yielding mechanism:

For the safety of the operator, the blade is equipped with a hydraulic blade yielding mechanism, that is based on a pressure accumulator. The yielding mechanism is based on hydraulic pressure accumulator. The tilting also protects the blade edge. The blade release system allows the blades to tilt and move moves slightly also during normal use; do not obstruct the tilting mechanism. The service of the yielding system and the pressure accumulator must be left to professional maintenance.

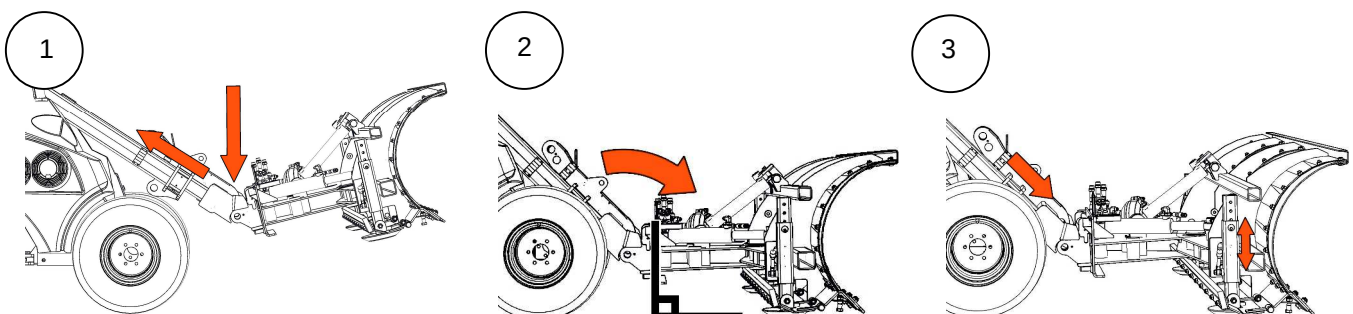
The blades may tilt only while driving forward. When reversing with the loader the blade sections are rigid, and therefore hitting an obstacle while reversing may lead to a sudden stop.

6.1. Operating position of the snow blade

For the best performance of the snow blade, it is important that the snow blade and its supporting skids are set up correctly. It must also be noted that the blade should not be pressed against the ground. If the blade is pressed hard against the ground, the manoeuvrability of the loader impaires as the front tyres lift off the ground.

Set the attachment to a position where the blade edge is in contact eith the ground, while the quick attach plate is set to upright position:

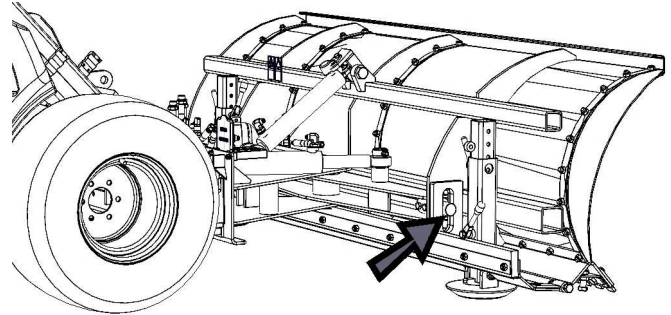
1. Retract the telescopic boom completely and lower the boom to its lowest position.
2. Tilt the attachment so that the quick attach plate is in uproght position.
3. Use the telescopic boom as necessary, so that both the slide skids and the blade edge ar in contact with the ground. Adjust the skids as necessary (section 6.3).
4. When set correctly, the play of the floating attachment braket (sections 6.2.) is in the middle position, when the snow blade is adjusted on even surface.



During operation, lift the boom without tilting the attachment. When lifting the blade using only the lifting and lowering movements of the boom, the most efficient operating position is maintained. If the blade is tilted separately, the correct position must be reset as shown above, and this will make working with the blade less efficient.

6.2. Floating attachment bracket

The floating attachment bracket allows the blade to follow the ground surface. For the best operation of the floating, the blade must set on even surface so that there is even play at the bracket. The correct position is set with the position of the boom and the supporting skids.



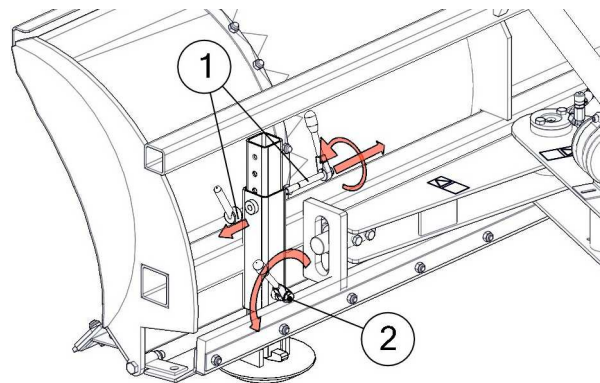
Boom floating of the loader:

The loader boom floating is not necessary, but it can be used to gain greater movements of the blade. Switch the boom floating of the loader on after the the blade is adjusted to correct position.

6.3. Adjusting the skids

The supporting skids help to achieve more even performance from the blade. They also lower the lateral force created by the blade, making it easier to control the loader. The height of the skids must be adjusted, if both the blade and the skids are not in contact with the ground, when the attachment bracket is in upright position. Set the blade as shown in 6.1 and set the skids against the ground.

1. Remove the fixing screw
2. Loosen the tightening screw and adjust the skid



6.4. Adjusting and replacing the blade

To prevent damaging the blade, the frame of the attachment must not be in contact with the ground. Adjust or replace the replaceable blade edges before the frame becomes in contact with the ground.

When adjusting or replacing the blade edges, keep the snow blade attached to the loader. Perform the task on even surface. Lift the attachment and prevent unintended movements of the blade by placing blocks of wood, for example, under the bracket, or by using suitable lifting equipment. The blade consists of two pieces of A35220.



The blade edges must be adjusted or replaced before the frame of the attachment becomes into contact with the ground.

There are no user adjustable components in the blade yielding system. If the system does not perform as intended, the pressure accumulator must be checked and adjusted at an authorized service point.

6.4.1. Rubber blade (optional extra)

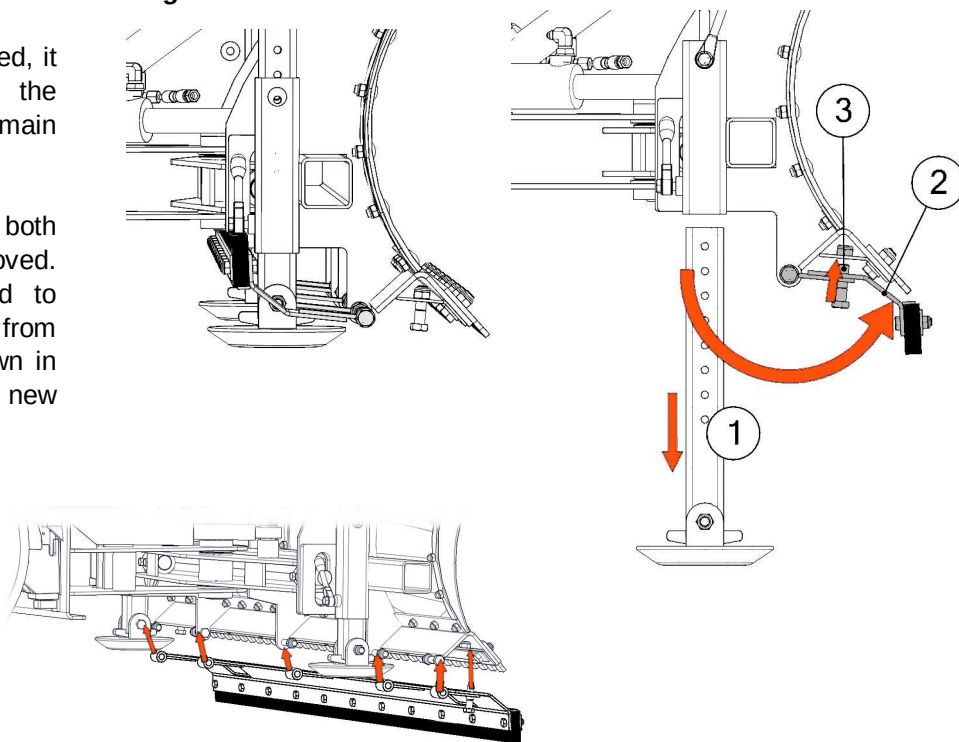
A rubber blade (A21497) is available for using the snow blade on more fragile surfaces, for example. Once the rubber blade is installed, it can be taken into use quickly as necessary. Part number for a spare rubber blade is A35186.

Putting the rubber into service and installing the blade

When the rubber blade is not used, it is turned backwards on the supporting skids, where it will remain ready to be used.

To take out the rubber blade, both supporting skids must be removed. After this, the blade is locked to service position with screws (3) from both ends of the blade, as shown in adjacent figure. Set the skids to new correct position.

When installing the rubber blade, it is not necessary to remove the supporting skids. rrotaa.



7. Maintenance and servicing

Continuous maintenance of the snow plow includes regular cleaning and lubrication of the equipment. The blade edge should be adjusted or replaced when needed.



Never go under a raised attachment or the loader boom. The loader boom will move down and the attachment can may be tilted even when the engine has been shut down. Ensure proper supportment of the attachment during all maintenance procedures. Use the boom supporting bracket provided with the loader and also additional means if necessary, or suitable lifting equipment.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a leak in the hydraulic system. Leaking hydraulic fluid may penetrate skin and cause serious injuries. The fluid is also harmful to the environment.



If any of the hydraulic components is damaged, the hose or component must be replaced. Because of the pressure accumulator in the hydraulic system of the blade yielding mechanism, only a professional service technician or mechanic familiar with hydraulic assemblies is allowed to service the system. Maintenance services and spare parts are available from Avant dealers and service points.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



7.1. Cleaning the equipment

You may use a pressure washer and a mild detergent to wash the equipment. Avoid spraying directly at the hydraulic components or using solvents when washing. In addition, mind the labels on the attachment. Lubricate the lubrication points after washing.

Clean the attachment thoroughly before long storage periods, Touch up the paint where needed to prevent rust damage and lubricate the lubricating points. The snow plow should be stored in a position where the hydraulic cylinders are completely retracted. Protect the visible part of the cylinder shafts by applying a small amount of oil to them. Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example. Ensure the stability of the equipment during storage periods.

7.2. Lubrication

There are several lubricating points on the snow plow, which can be lubricated with a standard grease press. Lubricating points are located at the floating attachment bracket, the hinge between the blades and the hinges at the tilting lower blade sections.



Lubricant must be added about 1 to 3 times a year. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

7.3. Pressure accumulator

The pressure accumulator is connected fixedly to the rod side port of the cylinder in the yielding mechanism. The system is precharged and ready to be used. The operation of the system cannot be adjusted without special tools. The charge pressure of the accumulator must be checked, if the yielding mechanism does not work as intended (too little force). Contact service. The accumulator can be charged only with nitrogen.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller: AVANT TECNO OY
Osoite / Adress / Address / Adresse: Ylötie 1
 33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

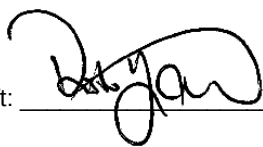
We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Kuvaus: Avant-kuormainten kanssa käytettävä lumiaura
Beskrivning: Snöplog; arbetsredskap för Avant lastare
Description: Snow plow; attachment for Avant loaders
Beschreibung: Schneeschild; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant	600- 700 series
Lumiaura 2200 Snöplog 2200 Snow blade 2200 Schneeschild 2200	A21493

Allekirjoitus / Namnteckning / Signature / Unterschrift:  9.12.2011 Ylöjärvi, Finland

Risto Käkälä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





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